Yuliang Peng

ypeng12@umd.edu | ypeng12.github.io/Mypage/ | linkedin.com/in/ypeng12 |240-886-9716

EDUCATION

University of Maryland, College Park

- Master of Science in Applied Machine Learning **GPA: 3.7/4.0 | May 2027**
- Bachelor of Science in Computer Science **GPA: 3.6/4.0 | May 2024**
- Relevant Courses: LLM Methods & Applications, LLM Systems, Cloud Computing, Computer Vision, Deep Learning

WORK EXPERIENCE

ModeSens - *Software Engineer* | Remote, USA | Apr 2025 - Sep 2025

- Built backend services for product ingestion across **30+** global retailers using automated pipelines and serverless workflows.
- Engineered production data pipelines with **CI/CD automation**, reducing failures by **20%** and improving reliability.
- Designed scalable Python and AWS S3 workflows for feature extraction and data standardization across retailer feeds.
- Built an internal **image-matching pipeline** with perceptual hashing, SSIM, and Selenium, reducing manual effort by **80%** and improving recommendation accuracy by **25%**.

Bank of China - *Software Engineer* | Los Angeles | July 2024 - Apr 2025

- Built backend services for financial transaction **monitoring systems**, supporting large-scale risk and compliance workflows.
- Engineered **data pipelines** for transaction processing and feature extraction, enabling analytics and fraud detection.
- Resolved **70+** production issues, reducing blocking incidents by **20%** and improving system stability.

Air China - *Software Engineer Intern* | Chengdu, China | July 2023 - Apr 2024

- **Automated system backups** across Linux environments using Unix shell scripts, reducing manual workload by **25%.**
- Revamped the secure **login system** with mobile QR code scanning, streamlining user identification. (adopted in 2024)

PUBLICATION / RESEARCH

International Cargo Charter Operations and Preighter Performance | Transport Policy, Vol. 184, Article 104173, 2026

- Co-authored an in-press Transport Policy paper and built Python workflows to process CAAC flight approval records, compute route-network metrics, and visualize cargo charter/preighter evolution during COVID-19. DOI: 10.1016/j.tranpol.2026.104173.

University of Maryland, College Park - *Research Assistant* | Nov 2025 - Apr 2026

- Designed and optimized **GPU-accelerated** spatiotemporal embeddings for high-rate **Event Camera** streams, enabling efficient ML inference via fused CUDA kernels and SIMD-optimized pipelines.

PROJECT

TabIt - AI-Powered Chrome Extension | Google Chrome Extension

- Built an intelligent browser tab management system that automatically groups tabs using semantic similarity on tab embeddings.
- Engineered a **real-time clustering pipeline** with Web Worker parallelization, supporting **100+** tabs with sub-**100ms latency**.
- Designed incremental clustering and persistent client-side state for efficient updates and seamless session restoration.

Face Verification System | Machine Learning | GitHub Code

- Built an **embedding-based** face verification system on LFW, integrating pairwise identity matching, similarity scoring, calibrated confidence estimation, and threshold-based decisions.
- Developed a **reproducible** evaluation pipeline with deterministic pair generation, ROC-based threshold calibration, confusion-matrix analysis, tracked runs, and error-slice diagnostics.
- Packaged the verifier as a **Dockerized CLI** inference system with automated tests, latency profiling, and concurrent load testing for deployment-oriented evaluation.

Cloud-Edge Collaborative Intelligent Agent Routing System | Google Hackathon

- Built a multi-level routing system for **edge and cloud LLM inference**, reducing **latency** and improving **time-to-first-token**.
- Developed **intent-based routing** to select between local models and external APIs, achieving **>95%** local processing.
- Implemented a post-processing framework to **mitigate hallucination** across multi-step tool calls, improving **response consistency** and system robustness.

SKILLS

ML / AI: PyTorch, TensorFlow, CUDA, Hugging Face, QLoRA, PEFT, NumPy, Pandas
Systems & MLOps: Kubernetes, Kafka, Prometheus, Spark, AWS, GCP, Azure, Docker, PostgreSQL, Redis, Elasticsearch
Languages : Python, C++, C, Rust, Java, JavaScript, English (Fluent), Chinese (Native), Japanese (Native, N1)